B. Fun Permutation

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 900 | Tags: constructive algorithms, math, number theory |
https://codeforces.com/problemset/problem/2137/B

You are given a permutation* p of size n.

Your task is to find a permutation q of size n such that GCD†$(p_i+q_i, p_{i+1}+q_{i+1}) \geq 3$ for all $1 \leq i$ lt n. In other words, the greatest common divisor of the sum of any two adjacent positions should be at least 3.

It can be shown that this is always possible.

*A permutation of length m is an array consisting of m distinct integers from 1 to m in arbitrary order. For example, [2,3,1,5,4] is a permutation, but [1,2,2] is not a permutation (2 appears twice in the array), and [1,3,4] is also not a permutation (m=3 but there is 4 in the array).

†gcd(x, y) denotes the greatest common divisor (GCD) of integers x and y.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10^4$). The description of the test cases follows.

The first line of each test case contains an integer n ($2 \leq n \leq 2 \cdot 10^5$).

The second line contains n integers $p_1,p_2,\ldots,p_n$ ($1 \leq p_i \leq n$).

It is guaranteed that the given array forms a permutation.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output the permutation q on a new line. If there are multiple possible answers, you may output any.

Example

Input
3 3 1 3 2 5 5 1 2 4 3 7 6 7 1 5 4 3 2

Output
2 3 1 4 5 1 2 3 2 1 3 7 5 6 4

Note

In the first test case, GCD(1+2,3+3)=3≥ 3 and GCD(3+3,2+1)=3≥3, so the output is correct.